



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 wide notification number F-9-12/2001-U-3 of Government of India)

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Computer Networks/0705210503

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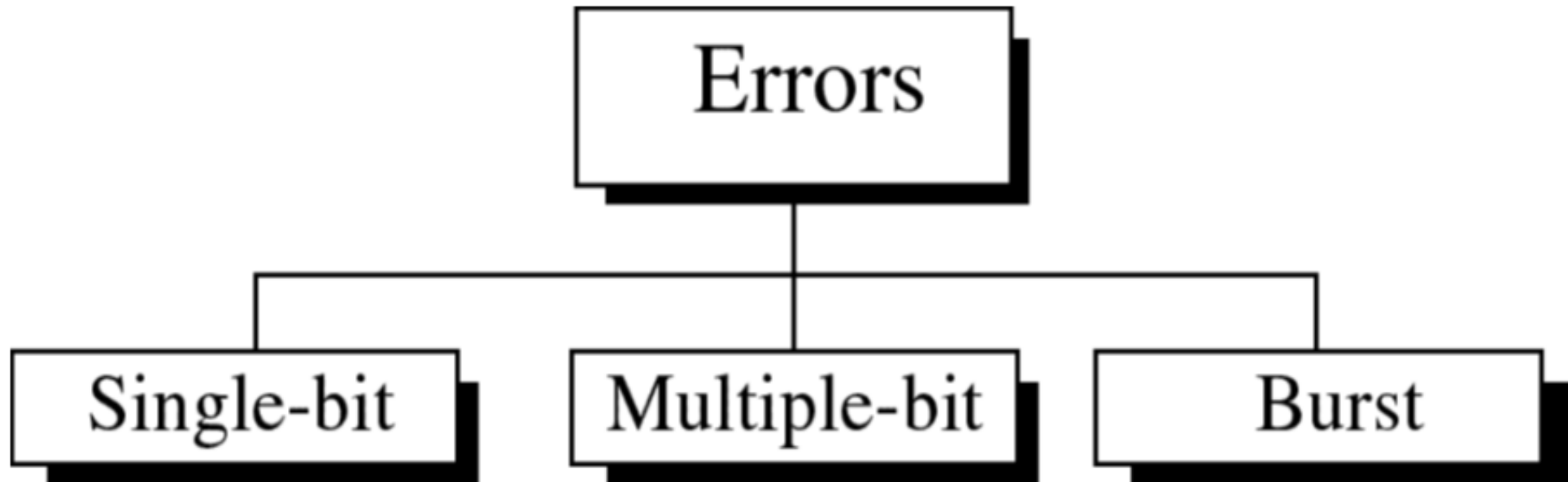


Error Control

- Networks must be able to transfer data from one device to another with complete accuracy.
- Data can be corrupted during transmission.
- For reliable communication, errors must be detected and corrected.
- Error control can be divided into two general categories:
 1. Error Detection
 2. Error Correction
- **Error detection and Error correction** are implemented either at the **data link layer** or the **transport layer** of the OSI model.

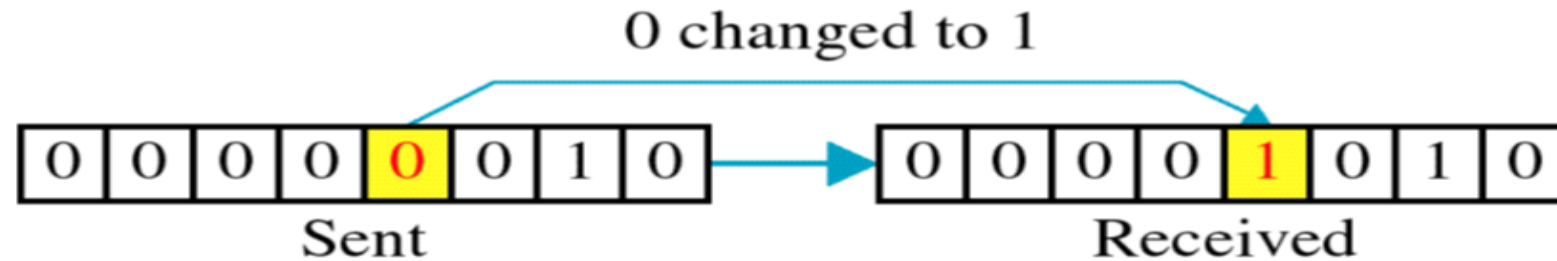


Types of Errors



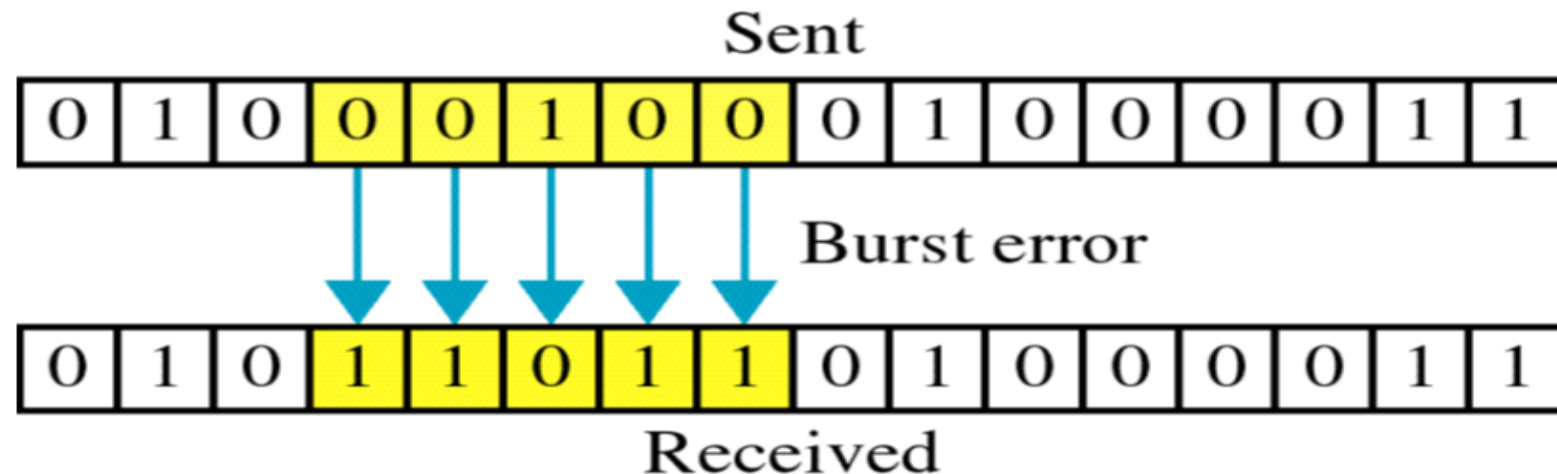


Single-bit error:



Single-bit errors are the least likely type of errors in serial data transmission because the noise must have a very short duration, which is very rare. However, this kind of error can happen in parallel transmission.

Burst error:



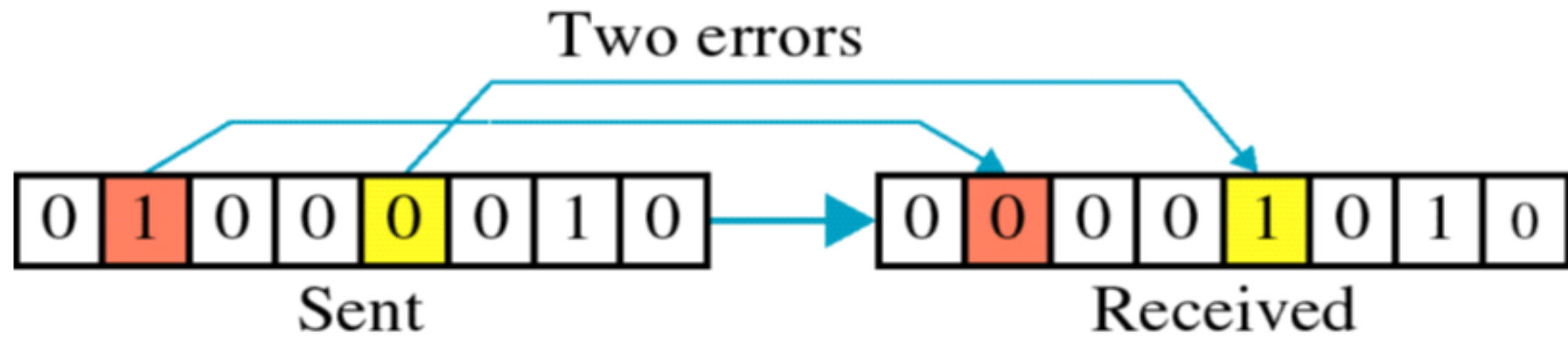


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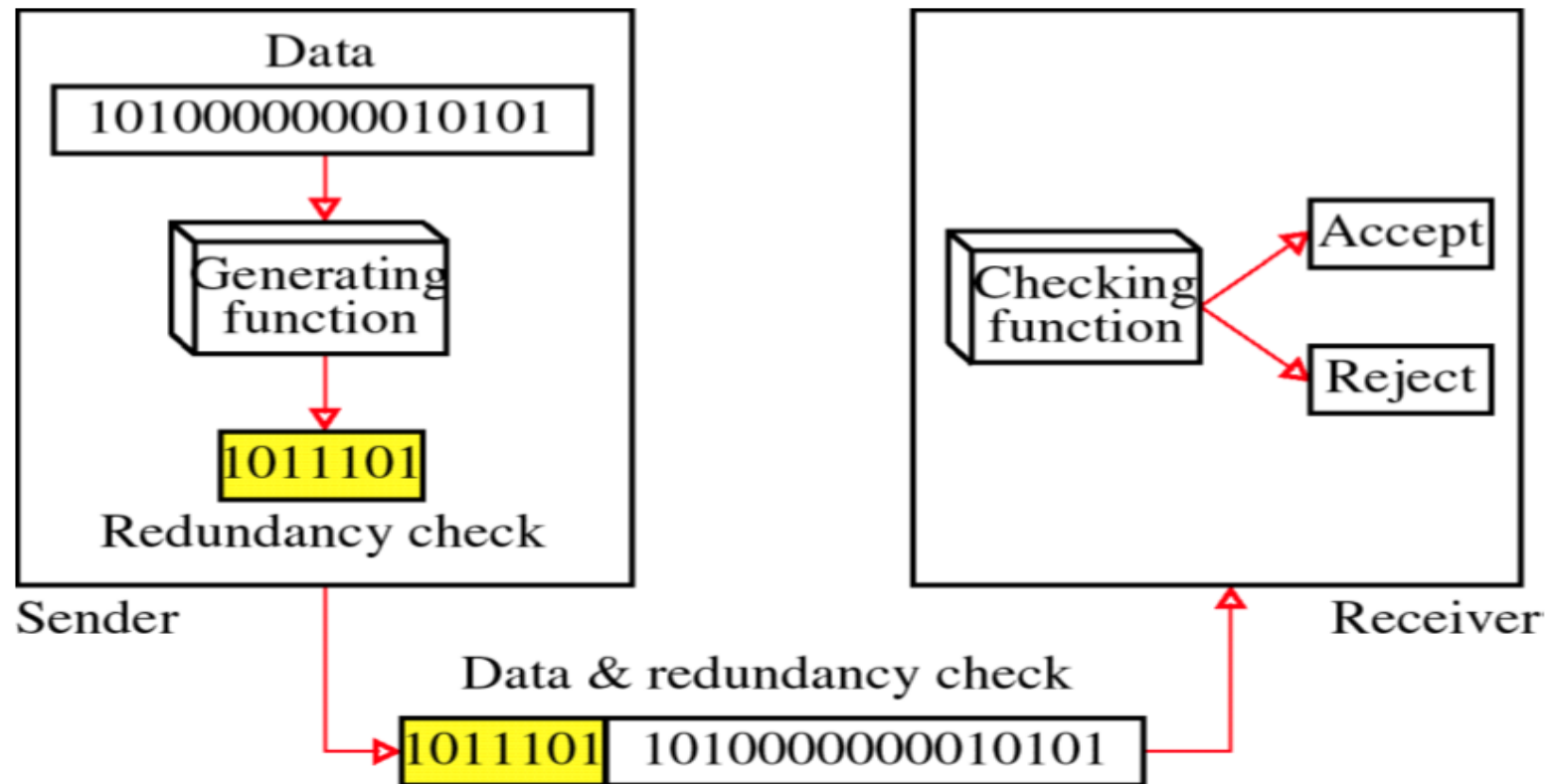


- The term **burst error** means that two or more bits in the data unit have changed from 1 to 0 or from 0 to 1.
- **Burst errors do not necessarily mean that the errors occur in consecutive bits;** the length of the burst is measured from the first corrupted bit to the last corrupted bit. Some bits in between may not have been corrupted.
- **Burst error is most likely to happen in serial transmission** since the duration of noise is normally longer than the duration of a bit. The number of bits affected depends on the data rate and duration of the noise.



Error detection

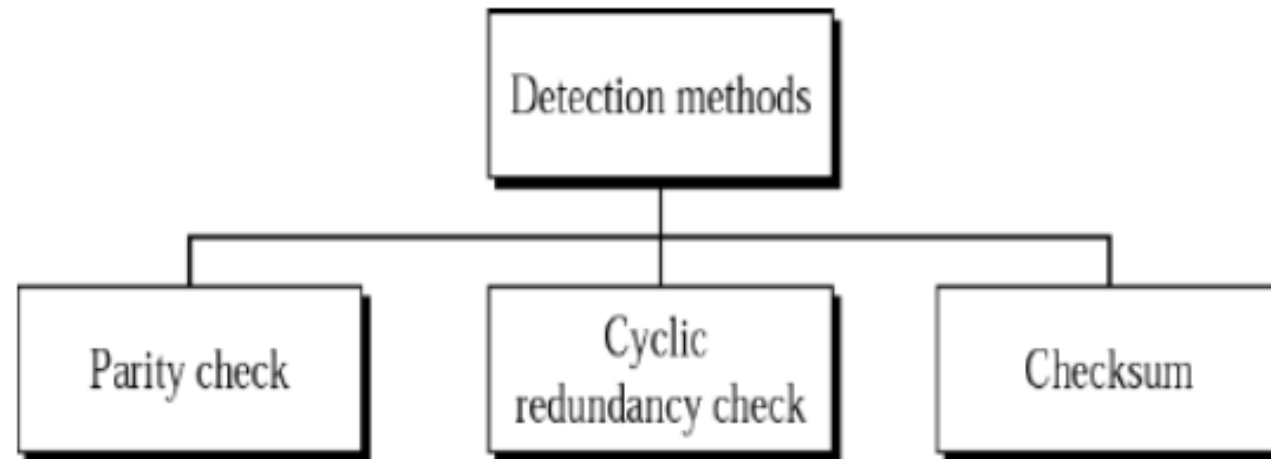
- Error detection means deciding whether the received data is correct or not without having a copy of the original message.
- Error detection **uses the concept of redundancy, which means** adding extra bits for detecting errors at the destination.





ERROR DETECTION TECHNIQUES / METHODS

The basic idea behind any error detection scheme is to add additional information to a frame that can be used to determine if errors have been introduced.





PARITY CHECK

- One bit, called parity bit is added to every data unit so that the total number of 1's in the data unit becomes even (or) odd.
- The source then transmits this data via a link, and bits are checked and verified at the destination.
- Data is considered accurate if the number of bits (even or odd) matches the number transmitted from the source.
- This technique is the most common and least complex method.

1. *Even parity* – Maintain even number of 1s

E.g., 1011 → 1011 **1**

2. *Odd parity* – Maintain odd number of 1s

E.g., 1011 → 1011 **0**



Hamming Code

- Hamming code is an error-correcting code used for correcting transmission errors in synchronous data streams.
- The Hamming code will correct only single-bit errors.
- It cannot correct multiple-bit errors.
- Hamming bits also sometimes called as error bits are inserted in to a character at random manner.

Parity bits: The bit which is appended to the original data of binary bits so that the total number of 1s is even or odd.

Even parity: To check for even parity, if the total number of 1s is even, then the value of the parity bit is 0. If the total number of 1s occurrences is odd, then the value of the parity bit is 1.

Odd Parity: To check for odd parity, if the total number of 1s is even, then the value of the parity bit is 1. If the total number of 1s is odd, then the value of the parity bit is 0.



The combination of data bits (***d bits***) and Hamming bits (***r bits***) called as **Hamming code** (***d+r bits***).

Number of redundancy bits (Hamming bits) '**r**' needed:

Let data bits = ***d***

Redundancy bits = ***r***

Total message sent = ***d+r***

The value of '**r**' must satisfy the following relation:

$$2^r \geq d+r+1$$

Algorithm of Hamming code:

- Information of '***d***' bits is added to the redundant bits '***r***' to form ***d+r***.
- The location of each of the (***d+r***) digits is assigned a decimal value.
- The '***r***' bits are placed in the positions **1,2,..... 2^{k-1}** .
- At the receiving end, the parity bits are recalculated. The decimal value of the parity bits determines the position of an error.



Suppose the original data is **1010** which is to be sent.

Total number of data bits 'd' = 4

Number of redundant bits r: $2^r \geq d+r+1$

$$2^r \geq 4+r+1$$

Therefore, the value of ***r*** is **3** that satisfies the above relation.

Total number of bits = $d+r$ = $4+3 = 7$;



The hamming method uses the extra parity bits to allow the identification of a single-bit error.

- **Step 1** First write the bit positions starting from 1 in a binary form (1, 10, 11, 100, etc.)
- **Step 2** Mark all the bit positions that are powers of two as parity bits (1, 2, 4, 8, 16, 32, 64, etc.)
- **Step 3** All other bit positions are for the data to be encoded using (3, 5, 6, 7, 9, 10 and 11, etc.)



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Position	R8	R4	R2	R1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1

R1 -> 1,3,5,7,9,11

R2 -> 2,3,6,7,10,11

R3 -> 4,5,6,7

R4 -> 8,9,10,11



Determining The Position of Redundant Bits

A redundancy bits are placed at positions that correspond to the power of 2. As in the above example:

- The number of data bits = 7
- The number of redundant bits = 4
- The total number of bits = $7+4=11$
- The redundant bits are placed at positions corresponding to power of 2 that is 1, 2, 4, and 8





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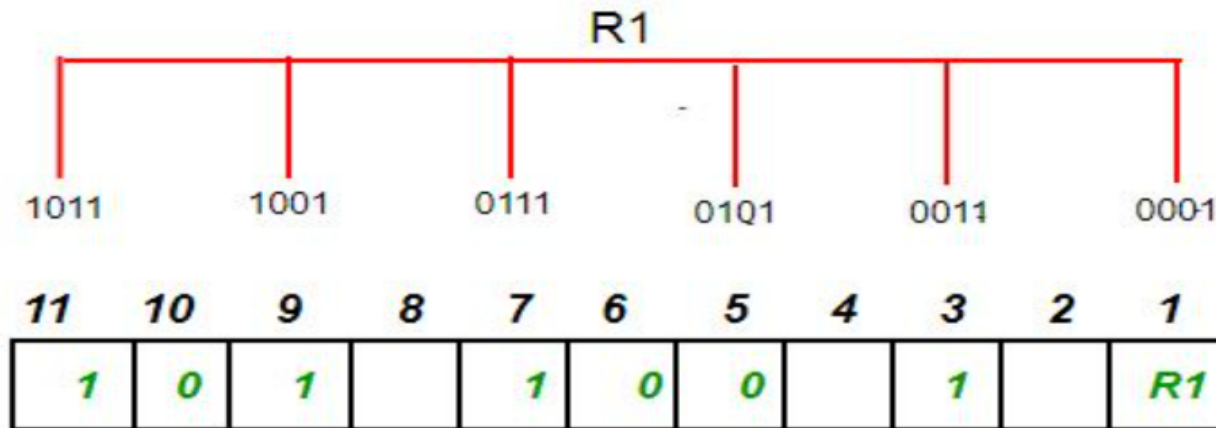
- Suppose the data to be transmitted is 1011001 from sender to receiver, the bits will be placed as follows:

11	10	9	8	7	6	5	4	3	2	1
1	0	1	R8	1	0	0	R4	1	R2	R1



Determining The Parity Bits According to Even Parity

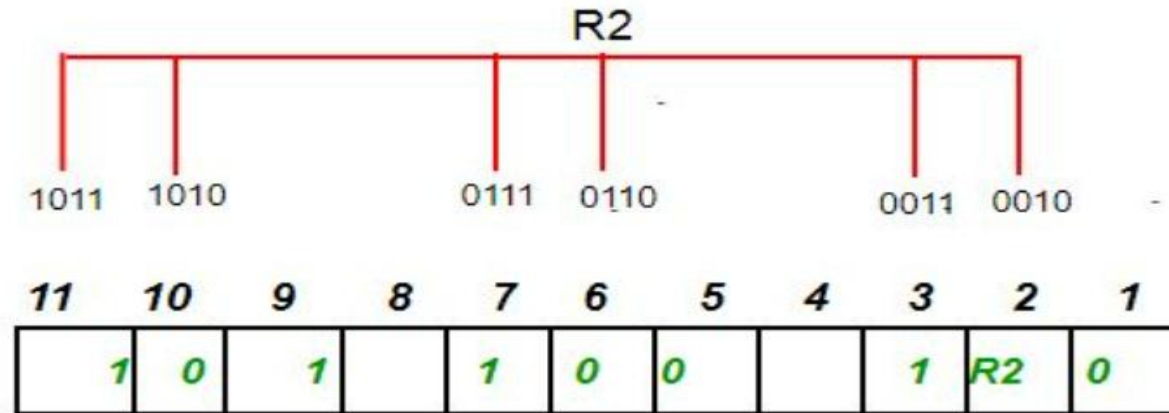
- R1 bit is calculated using parity check at all the bits positions whose binary representation includes a 1 in the least significant position. R1: bits 1, 3, 5, 7, 9, 11



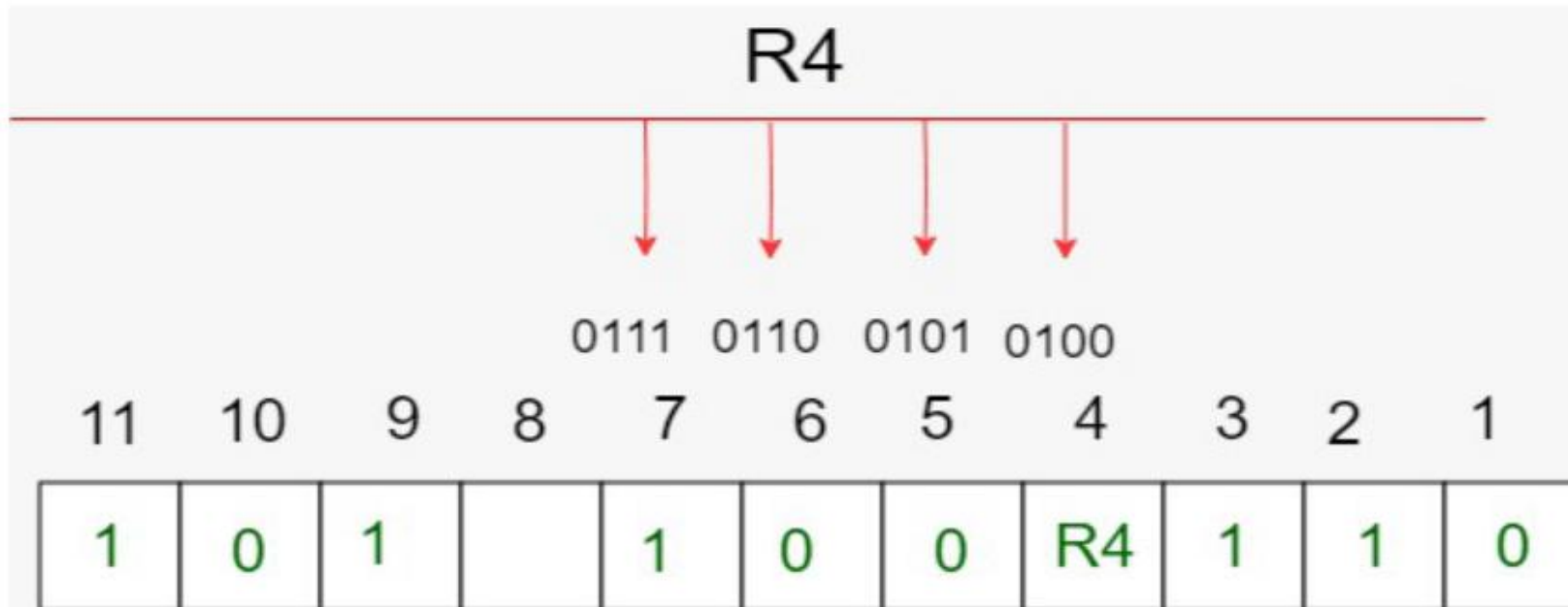
- To find the redundant bit R1, we check for even parity. Since the total number of 1's in all the bit positions corresponding to R1 is an even number. So, the value of R1 (parity bit's value) = 0.



- R2 bit is calculated using parity check at all the bits positions whose binary representation includes a 1 in the second position from the least significant bit.
R2: bits 2,3,6,7,10,11



- To find the redundant bit R2, we check for even parity. Since the total number of 1's in all the bit positions corresponding to R2 is odd the value of R2 (parity bit's value) = 1
- R4 bit is calculated using parity check at all the bits positions whose binary representation includes a 1 in the third position from the least significant bit.
R4: bits 4, 5, 6, 7



To find the redundant bit R4, we check for even parity. Since the total number of 1's in all the bit positions corresponding to R4 is odd so the value of R4 (parity bit's value) = 1

R8 bit is calculated using parity check at all the bits positions whose binary representation includes a 1 in the fourth position from the least significant bit.

R8: bit 8,9,10,11

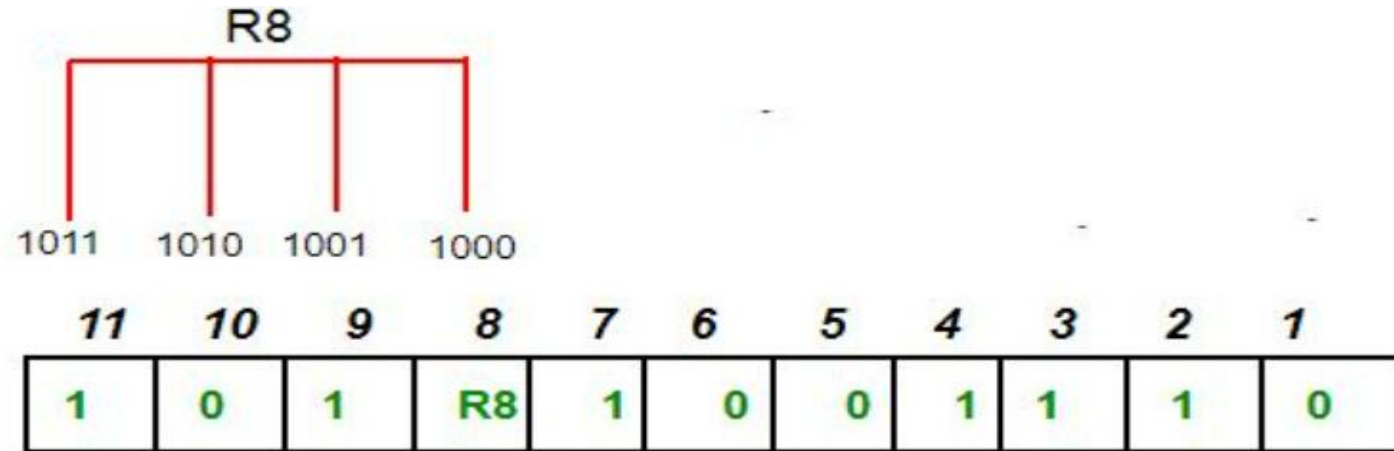


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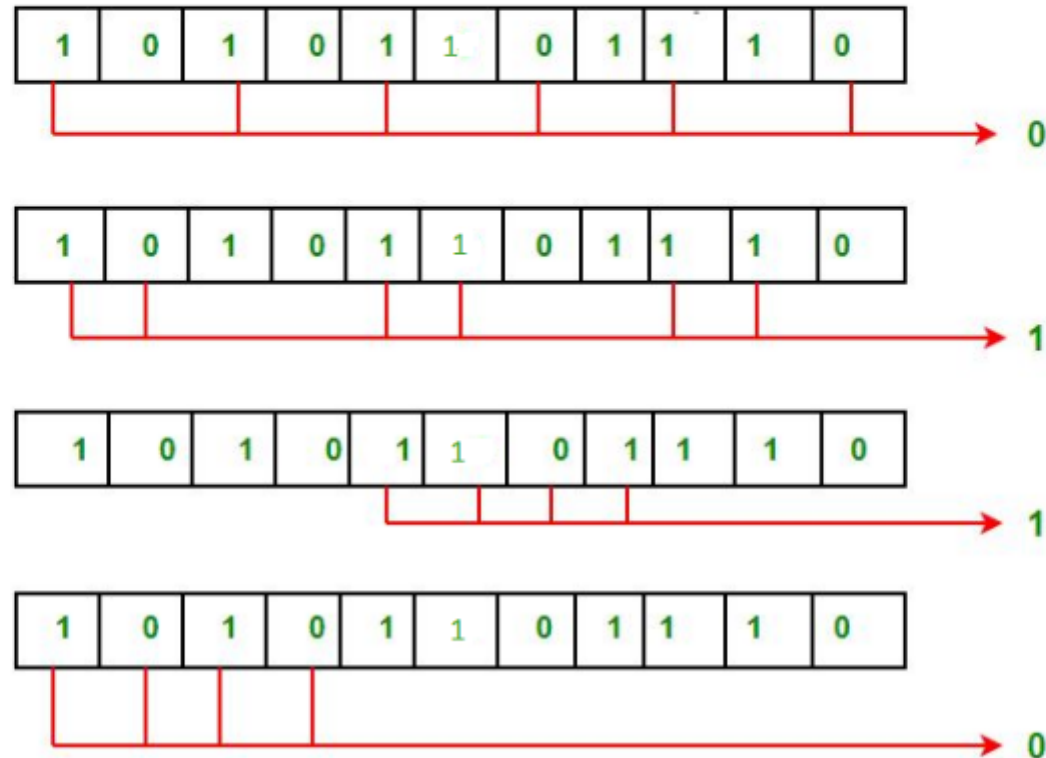
To find the redundant bit R8, we check for even parity. Since the total number of 1's in all the bit positions corresponding to R8 is an even number the value of R8 (parity bit's value) = 0. Thus, the data transferred is:

11	10	9	8	7	6	5	4	3	2	1
1	0	1	0	1	0	0	1	1	1	0



Error Detection and Correction

Suppose in the above example the 6th bit is changed from 0 to 1 during data transmission, then it gives new parity values in the binary number:





For all the parity bits we will check the number of 1's in their respective bit positions.

- For R1: bits 1, 3, 5, 7, 9, 11. We can see that the number of 1's in these bit positions are 4 and that's even so we get a 0 for this.
- For R2: bits 2,3,6,7,10,11 . We can see that the number of 1's in these bit positions are 5 and that's odd so we get a 1 for this.
- For R4: bits 4, 5, 6, 7 . We can see that the number of 1's in these bit positions are 3 and that's odd so we get a 1 for this.
- For R8: bit 8,9,10,11 . We can see that the number of 1's in these bit positions are 2 and that's even so we get a 0 for this.
- The bits give the binary number 0110 whose decimal representation is 6.

Thus, bit 6 contains an error. To correct the error the 6th bit is changed from 1 to 0.



CYCLIC REDUNDANCY CHECK

- Cyclic codes refers to encoding messages by adding a fixed-length check value.
- CRCs are popular because they are simple to implement, easy to analyze mathematically and particularly good at detecting common errors caused in transmission channels.

Steps Involved :

- Consider the original message (dataword) as $M(x)$ consisting of 'k' bits and the divisor as $C(x)$ consists of 'n+1' bits.
- The original message $M(x)$ is appended by 'n' bits of zero's. Let us call this zero-extended message as $T(x)$.
- Divide $T(x)$ by $C(x)$ and find the remainder.
- The division operation is performed using XOR operation.
- The resultant remainder is appended to the original message $M(x)$ as CRC and sent by the sender(codeword).

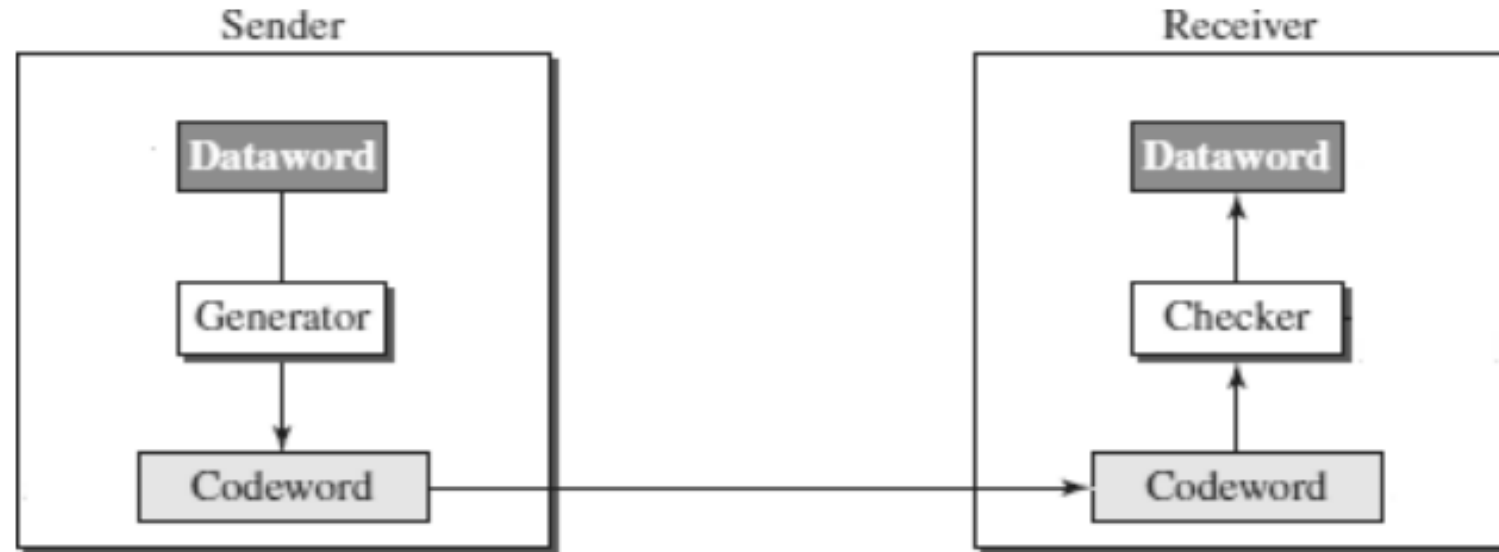


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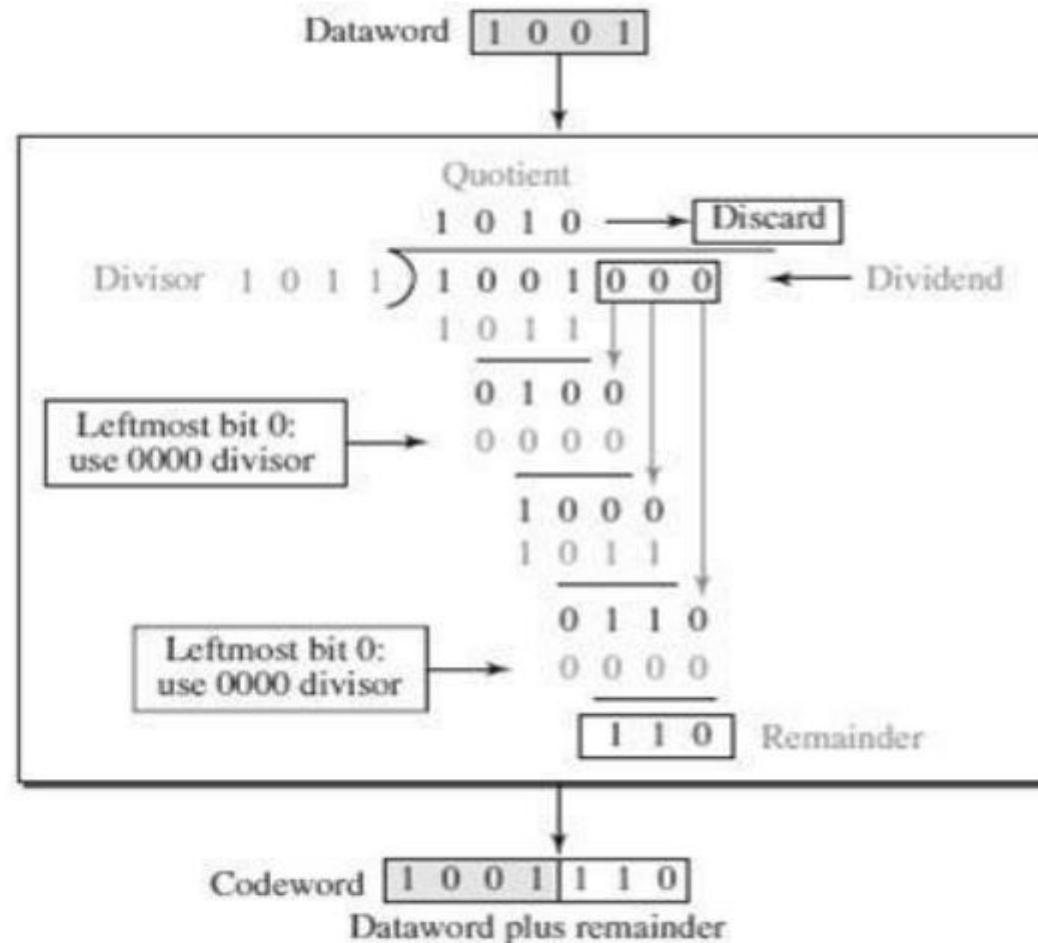


Example 1:

- Consider the Dataword / Message $M(x) = 1001$
- Divisor $C(x) = 1011$ ($n+1=4$)
- Appending 'n' zeros to the original Message $M(x)$.
- The resultant message is called $T(x) = 1001$ **000**. (here $n=3$)
- Divide $T(x)$ by the divisor $C(x)$ using XOR operation.



Sender Side :



Note:

Multiply: AND
Subtract: XOR



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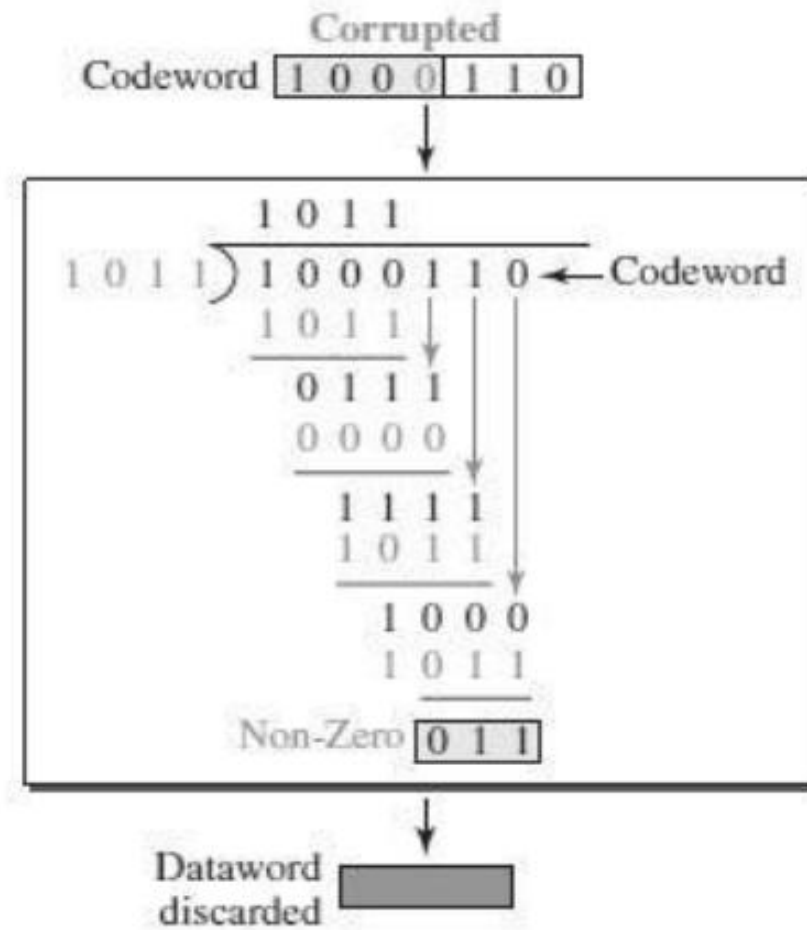
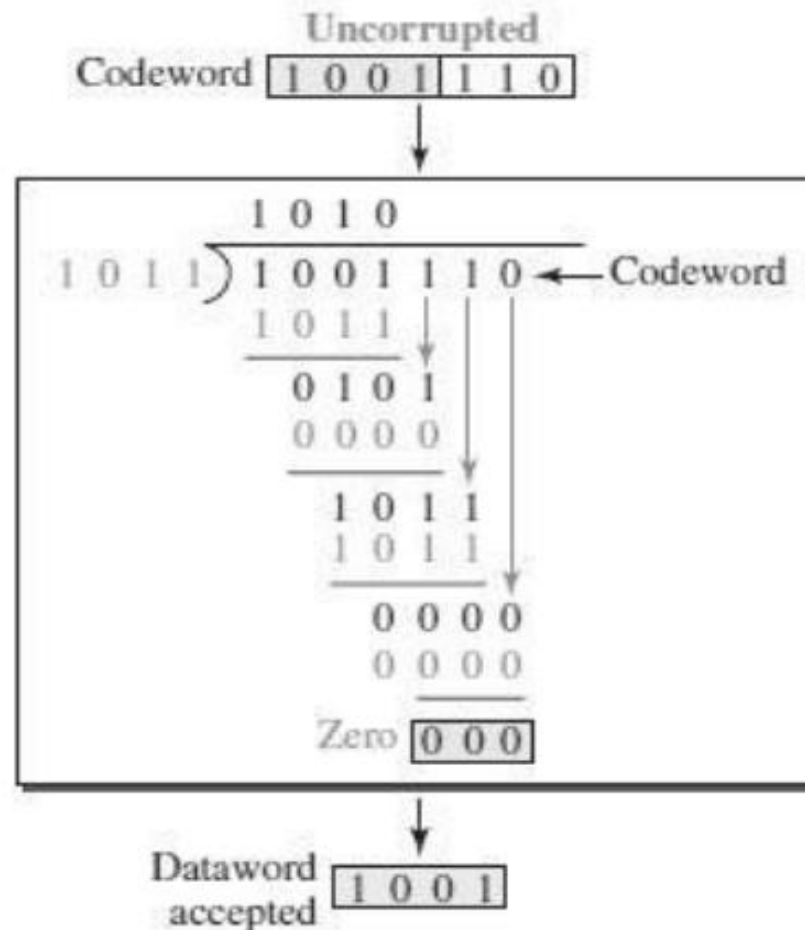
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Receiver Side:

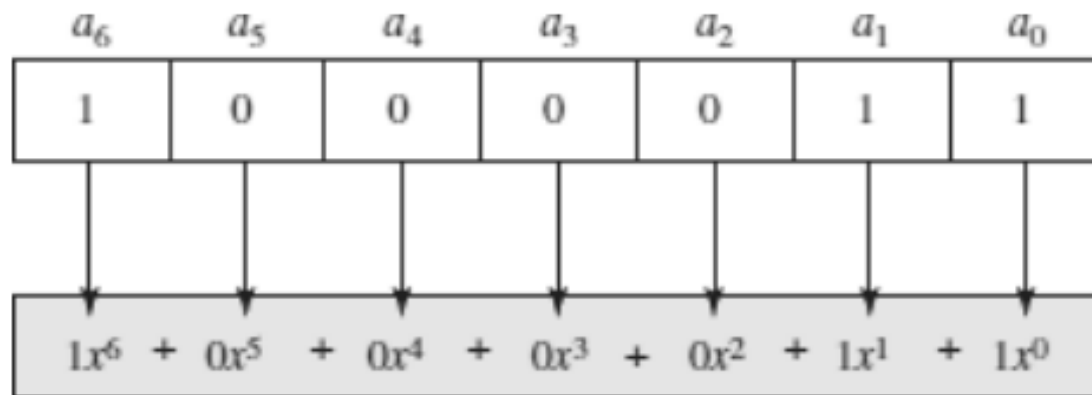
(For Both Case – Without Error and With Error)



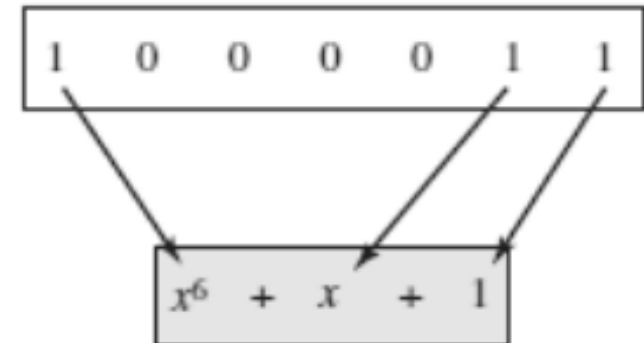


Polynomials

- A pattern of 0s and 1s can be represented as a **polynomial** with coefficients of 0 and 1.
- The power of each term shows the position of the bit; the coefficient shows the value of the bit.



a. Binary pattern and polynomial



b. Short form